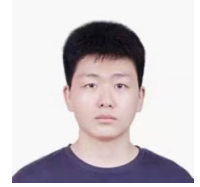


Ranjun Peng

B.Sc. in Artificial Intelligence | Research interests: LLMs & AI for Software Engineering

+86 18310886970 | 1230024861@student.must.edu.mo | Macau, China

Highlights: 4 publications, including a sole first-author paper under review at ACM TOSEM (CCF-A).



Education

Macau University of Science and Technology

Sep 2023 – Jun 2027 (expected)

B.Sc. in Artificial Intelligence | GPA: 3.3/4.0

Core courses: Machine Learning, Deep Learning, Data Mining, Computer Vision, Data Structures & Algorithms, Database Systems.

Publications & Research

Understanding Software Defect Prediction: A Large-scale Empirical Study Across Uncertainty Quantification and Performance Evaluation Sole first author | ACM TOSEM (CCF-A) – under review

- Independently designed and ran a large-scale empirical study: 16 representative ML classifiers across 36 within-project and 32 cross-project datasets, with 5 uncertainty / 6 performance / 3 calibration metrics over 30 seeds; built the full pipeline (COD-clustering model selection, training/calibration, correlation analysis) and showed probability-based uncertainty is highly context-dependent and transfers poorly across projects.

Data Feature-Constrained GAN Data Generation for NIR Spectral Predictive Model Calibration

First author | 36th Chinese Process Control Conference (CPCC 2025)

- Proposed DFC-GAN, embedding the PCA principal-component covariance matrix into the generator loss as a statistical constraint; implemented the full architecture and custom loss, cutting downstream SVR/LSTM prediction RMSE by up to 50% on a public pharmaceutical NIR dataset.

Dictionary Learning-Based Knowledge Fusion ESN Hybrid Modeling

Second author | Industrial &

Engineering Chemistry Research (ACS), published

- Co-designed and implemented the core ML algorithms (dictionary learning with sparse coding, knowledge-constrained ESN reservoir optimization); reproduced and trained all baselines (LSTM, SVR, Transformer-based Autoformer) on simulated and real chemical-plant data. DOI: [10.1021/acs.iecr.5c01050](https://doi.org/10.1021/acs.iecr.5c01050)

Towards Hybrid Modeling of Process Industries with Dynamic Mechanism and Data Augmentation

Second author | 44th Chinese Control Conference (CCC 2025)

- Co-developed an information-entropy data augmentation method (KDE with L_1 sparsity) and an Elastic Net error-compensation model; benchmarked against MLP/SVR/LSTM on CSTR simulation (MAE/RMSE/ R^2).

Technical Projects

AI Closet – Web-based Smart Wardrobe

Personal Project | 2026

Stack: Next.js / TypeScript / React / OpenAI Vision / Vitest

- Built a multimodal clothing-understanding pipeline using a Vision LLM to parse garment images into structured tags (category/color/style/occasion), with a preference-feedback re-ranking algorithm and offline fallback.

Soul Saber – Interactive AI Lightsaber System

Course/Personal Project | 2026

Stack: ESP32-S3 / Arduino C++ / Python / MPU6050 / NeoPixel

- On-device real-time gesture recognition from 6-axis IMU data (latency under 50ms), plus a low-latency ASR → LLM → TTS voice-interaction pipeline via a Python bridge for edge-cloud inference.

Internship

Bluestar Intelligent Cloud (Shandong) Technology Co., Ltd. – Beijing Branch

Jun 2025 – Aug 2025

Technical Intern, Production Digitalization Dept. (Industrial Software Development)

- Contributed to industrial software development for smart-manufacturing scenarios: requirement analysis, feature development, integration testing, and documentation.

Technical Skills

Programming Python (proficient), C/C++ (embedded), SQL.

LLMs Transformer/GPT/LLaMA architectures; SFT/RLHF/DPO fine-tuning; Vision-LLM integration and ASR → LLM → TTS pipelines.

ML / Data Scikit-learn (large-scale classification/calibration/evaluation), PyTorch (training & debugging).

Frameworks PyTorch, HuggingFace Transformers, NumPy/Pandas/Matplotlib, Git, Linux, Arduino/ESP-IDF.

Languages English (fluent, English-taught program), Mandarin (native).